

Notes on "Arrows, Structures and Functors: Categorical Imperative"

1 Learning to think with arrows

1.1 Epimorphisms and Monomorphisms

- A function $f : A \rightarrow B$, with **domain** A and **codomain** B , is **onto** or **surjective** if every element b (or $f(a)$) in the codomain B is mapped by at least one element a in the domain A . So for instance, $f : \mathbb{Z} \rightarrow \mathbb{Z} : n \mapsto n + 1$ is onto. Another example of such is the identity mapping, e.g. $id_A : A \rightarrow A$. An example of a function which isn't onto is for instance $f : \mathbb{Z} \rightarrow \mathbb{Z} : n \mapsto n^2$, since all negative integers in the codomain aren't mapped by any element in the domain.
- Two functions are said to be the same if they have the same domain and codomain, and yield the same value given a same argument.

Proposition 1.1.1

If a map $f : A \rightarrow B$ is onto, then whenever two maps $g, h : B \rightarrow C$ satisfy $g \circ f = h \circ f$, g and h must be equal.

Proposition 1.1.2

DEF. A function f is an **epimorphism** if $g \circ f = h \circ f$ always implies $g = h$.

Proposition 1.1.3

A function $f : A \rightarrow B$ is onto iff it is an epimorphism.